

QMHP-CoPro

Heavy-Fluxonium Erasure-Conversion Coprocessor Architecture
**v1.5.8f — Master Edition (Final) · Matched-Construction Branch Screen ·
Realistic-Location Cost Study · Consolidated Verification Register**

24 August 2026 · TRL-2 · Supersedes v1.5.8e (both builds), AMD-C r1-r6 and Master-rc1 · ERRATA-1 ER-1...ER-8 applied

EDITION, PROVENANCE AND ANCHORS (ER-1 applied)
This master edition consolidates the v1.5.8e release, amendment AMD-C r6 and errata ERRATA-1 into a single normative document with edit register E1-E20 applied inline. Dual provenance anchors: the OPERATIVE base is the 117-page v1.5.8e build, SHA-256 **d334cda822c07c212be1fac850a34debef5ea66ee45899e56b44e19154f86c6c** (T.12-T.15, S2.2/S2.3, modules S5.1-S5.6); the 109-page render, SHA-256 **83aeb3a2d8ebcb8f627989557350d9c172ad5bc20c3edc561b8a5e091cc4b658**, is PROVENANCE ONLY (E17). Archival material — complete S1 shot grids, executable listings, legacy Appendix-S tables — is carried by reference to the operative anchor. Nothing in this document is a hardware result. Every newly executed computation is equation-defined and re-executable; stochastic results declare seeds and conventions. Figures 1-4 are regenerated from the executed artifacts of the August 2026 verification cycle; each caption names its shipped data source (ER-2 applied). BUILD 2 (24 Aug 2026): release-candidate review fixes RB-1-RB-4 applied — requirements.lock made pip-valid; RUN.md anchors corrected to the E17 designation; T.13' split into convention-named artifacts (t13_distance_trend_release_hazard.json primary, ..._linear_sensitivity.json) with Figure 4 regenerated from the primary; RUN_MASTER.md and build_master_figures.py shipped so every figure regenerates from shipped artifacts. No numerical changes.

VERIFICATION STATE
Every load-bearing computational layer of the branch-screen result has independent confirmation across three evidence tiers — release execution (Tier A); reconstruction-level re-execution, 29/39 grid rows bit-exact with the remainder inside printed Wilson bands (Tier B); independent replication from the stated equations alone, two builds agreeing at all shared points (Tier C). Extension studies carry the graded status of §15. Status: FINAL — ready to freeze; the fixed-point digest of this artifact is written to the external release manifest.

Decision brief (E9)

Current state

Continue as a gated research program. The v1.5.8d branch screen is WITHDRAWN as decision-bearing evidence in either direction. Under the matched-construction row-wise map, Branch A and Branch C are statistically indistinguishable at $d = 3, 5, 7$ with no erasure-location credit ($\lambda_A/\lambda_C = 0.94 [0.88, 1.01], 0.92 [0.79, 1.08], 1.15 [0.97, 1.36]$); Branch A is lower at $\geq 95\%$ confidence at every distance under the ideal free-location bracket ($0.43 / 0.20 / 0.15$). Because Branch-A inputs are allocations and Branch C is a cross-device prior, neither branch is selected and neither is excluded: NO SEPARATION ESTABLISHED, superseding the v1.5.8d adverse screen (CONSTRUCTION-SENSITIVE).

Excluded claims — this edition does NOT support

Any processor-performance or fault-tolerance claim; any absolute p_L as a hardware prediction (QEC-MAP-PHENO is a model screen). The v1.5.8d conclusion that the erasure branch fails to earn its hardware — and equally any claim that Branch A has demonstrated advantage (the zero-location margin traces to lower charged load per unit time, not erasure conversion; premise-kill watch §13.2). Any confirmatory distance-trend claim from T.13' (EXPLORATORY, not pre-registered). Exact finite-sample ensemble counts as reproducibility targets (E19): the reported object is the rejection rate, $\approx 6.2\text{--}6.7\%$ across independent seed blocks (ER-8).

Top blocker	Status / evidence	Owner	Ordering gate (E10)
P6-E7 leakage-to-syndrome coupling	BLOCKING. Two LRU1 implementations differ $3.3\times$ at $d=3$ ($8.52e-3$ vs $2.840e-2$) and reverse the distance trend	B. Duncan (programme lead)	Adjudication rule and protocol frozen before comparative decoder data are opened
Driven sink survival $T_{\text{sink}}(n)$	BEYOND-DEMONSTRATED: 3-6 ms class vs $\approx 20\ \mu\text{s}$ An-like during-check comparator	B. Duncan	P6-E3 first cooldown; $q_{\text{miss}} \leq 5e-4$ re-optimised jointly with the cost-side gate (E3)
Full gate propagation (Stim-level)	Required before any absolute p_L is physical; tooling unblocked	B. Duncan	Frozen decoder/noise model before holdout (P7-pre)
Joint-device Branch-C control	Branch C is a cross-device literature prior	B. Duncan	Same-device, same-cooldown measured control before fabrication commit (§8.3)
Coupler/readout spectral replan	Readout moved to ≈ 4.30 GHz; legacy coupler plan not revalidated	B. Duncan	Regenerated spectral plan before tile-scale fabrication (§11)
Purcell filter $\kappa(\omega)$	34-36 dB stopband at dressed 3.395056 GHz mandatory under current anchors	B. Duncan	Measured $\kappa(\omega)$ around the 4.302 / 3.3951 GHz bands (P6-E2)

Owners are named on every blocker; targets are event-anchored ordering gates rather than calendar dates — deliberate at TRL-2, where the binding constraint is pre-registration order. Calendar dates attach when the corresponding campaign is funded.

Notation quick reference

Term	Meaning
Branch A / B / C	Half-flux heavy fluxonium ($ 0\rangle/ 2\rangle$ logical, $ 1\rangle$ sink) / integer-flux fluxonium / conventional-fluxonium null control
Pe, Pu	Located-erasure load per round ($1.543\text{e-}2$) / unlocated ledger per round (baseline $8.50\text{e-}3$; programme ceiling $1.0\text{e-}2$; headroom $1.50\text{e-}3$)
λ_L , HR	Logical hazard $-\ln(1-p_{L,\text{round}})/t_{\text{round}}$; hazard ratio λ_A/λ_C at equal wall clock ($t_A = 20\text{ }\mu\text{s}$; $t_C = 13.5\text{ }\mu\text{s}$ prior)
f, f*, p_induced	Location efficiency; break-even f vs C ($0.524 / 0.598$ at $d=3/5$); check-induced logical error per data qubit per round
twirlAD	Pauli-twirled amplitude damping, $pX=pY=pZ=\gamma/4$ (executed convention, both branches)
δ (E14)	Junction phase $\delta \equiv \phi - \phi_{\text{ext}}$; at $\phi_{\text{ext}} = \pi$, $\sin(\delta/2) = -\cos(\phi/2)$, parity-EVEN
F8	$\Gamma_P = \kappa(\omega_{\text{if}}) \cdot \langle f a i\rangle ^2$ — executable Purcell-weight definition (E18)
Tier A / B / C	Release execution / reconstruction-level re-execution / independent replication from stated equations (§15)

1. Executive decision and architecture state

Branch A remains physically interesting because the $\{|0\rangle, |2\rangle\}$ encoding is symmetry-protected against the first-order charge transition at half flux while $|2\rangle \rightarrow |1\rangle$ converts to a located event; the same symmetry forbids the direct logical charge drive and leaves an unlocated $|2\rangle \rightarrow |0\rangle$ bypass. Branch B has direct 2026 single-fluxonium erasure-conversion hardware evidence at integer flux, whose comparators also show check back-action can dominate the benefit. The $d=3$ baseline sets mid-circuit P6-E checks to zero because back-action is unproven; leakage-to-syndrome coupling during sink residence is the largest single uncertainty. The current program question: under an explicit row-wise error map, matched charging and equal elapsed time, does Branch A or B beat conventional Branch C after all branch-specific costs are charged? v1.5.8e executes the first matched model screen and returns NO SEPARATION. It does not substitute the screen for measured hardware.

1.1 What has been computationally closed

Domain	Frozen result
Half-flux spectrum / parity	$f01 = 0.175710013$, $f02 = 3.581724312$ GHz; $ n02 $ at numerical floor; $ \langle 0 \sin(\delta/2) 2\rangle = 0.295705232$ (6 per E14). Independently replicated to all nine figures, two Tier-C builds.
Pull convention	Eq.(7) χ is the negative of the physical cavity pull; equality/zeros unaffected; drive detuning and Stark/MIST direction use the physical sign.
Dressed logical-blind null (E13)	Production $Nq=10/Nph=12$: $f_r = 4.301974466$ GHz (two independent builds, nine figures). Spot check $Nq=14/Nph=25$: 4.301975383 GHz (the register value), $+0.92$ kHz, inside the 0.748 – 1.202 kHz band. Seed 4.314628047 GHz; correction -12.653 MHz. Pulls $-9.507/+8.721$ MHz; contrast 18.228 MHz; sink line 4.310696 GHz; dressed $f12 = 3.395056$ GHz (truncation-stable).
Fabrication null existence	$400/400 \sigma=2\%$ (EJ, EC, EL) samples retain a dressed null at the declared seed; per-device trim mandatory; root p5–p95 spread ≈ 4.19 – 4.41 GHz (draw-dependent at the MHz level).
Collision gate (E19, ER-8)	Sub-13-MHz ω_{24} rejection rate ≈ 6.2 – 6.7% : 6.66% $[6.12, 7.21]$ (20-seed block A, undeclared seeds, rc1) and 6.19% $[5.66, 6.72]$ (20-seed block B, seeds 1–20, shipped ensemble_20seed.json); dispersion consistent with $\text{Binomial}(400, p)$ in both. The release's $32/400 = 8.0\%$ is a $+1.1\sigma/+1.5\sigma$ single draw. Failing-device identities differ materially from a perturbative screen.
Static Purcell channel (E18, ER-4)	F8-exact weight $ \langle I,0 a 2,0\rangle ^2 = 0.01182198$ at $(12,18)/(14,25)$; production $(10,12)$ value $0.01182201 (+3e-8)$ — truncation-labelled. Bare-admixture overlap 0.011815 is a related, different object. Unfiltered $ 2\rangle \rightarrow 1\rangle$ lifetime $\approx 6.73/4.49 \mu\text{s}$ at $\kappa/2\pi = 2/3$ MHz; 34 – 36 dB stopband at 3.395056 GHz mandatory ($267\times/401\times$ to the 1.8 ms anchor, $\times 10$ allocation).
Finite-photon / Floquet / IQ	Direct pointer functional supersedes the n^3 proxy (0.021 – 0.058 photon ceiling withdrawn); leading high-level line $ 1\rangle + 9\gamma \rightarrow 14\rangle$ bounded inert in $n = 5$ – 8 ; worst ideal-Gaussian point SNR 6.91 , assignment error $5.10e-7$.
Row-wise map (F11)	$p_{\text{opp}} = n_{\text{data}} \cdot \text{alloc} / (n_{\text{opp}} \cdot q_{\text{per_event}})$ reproduces all fourteen $d=3$ rows to $<1e-9$; ledger re-aggregates to $8.500e-3$ exactly; independently reproduced.
Matched branch screen	NO SEPARATION at $d=3/5/7$ without location credit; A lower at $\geq 95\%$ at every distance with the ideal bracket; v1.5.8d result CONSTRUCTION-SENSITIVE.
Filter sharpness	Synthetic stopband 3.2056 – 3.6727 GHz, max 37.65 dB; 464.24 MHz transition to the lowest readout root (≥ 81.1 dB/GHz); paired same-device separation ≥ 761 MHz, median 905 MHz.

1.2 Open and hardware-bound

Filtered, driven sink lifetime $T_{\text{sink}}(n)$ under the actual P6-E tone including $e/|1\rangle$ destruction; MIST, TLS, QP/radiation response, array/package/coupler modes under the delivered spectrum; measured $\kappa(\omega)$ around 4.302 and 3.3951 GHz; P0d/P1 branch-resolved rates on one device and cooldown, reset ON/OFF; coupler-band redesign and EOL mapping; Branch-B optimization and ledger; the mechanism-resolved erasure-use study with measured check costs — T.12' (§14) executes the model-level variant; hardware measurement of $(f, p_{\text{induced}}, t_{\text{chk}}, \text{sink destruction, latency})$ remains decisive per P6-E6'.

2. Physical architecture and branch definitions

Branch A — half-flux heavy fluxonium, $(E_C, E_J, E_L)/h = (0.60, 5.72, 1.58)$ GHz at $\phi_{\text{ext}} = \pi$; logical $\{|0\rangle, |2\rangle\}$, monitored $|1\rangle$ sink. Charge-parity selection forbids $|0\rangle \leftrightarrow |2\rangle$ ($|n02|$ at floor); $|2\rangle \rightarrow |1\rangle$ is charge-allowed ($|n12| = 0.659915$) and is the located-erasure channel. Branch B — integer-flux fluxonium with 2026 external hardware evidence; own contract (Appendix M by reference). Branch C — conventional-fluxonium null control; a cross-device literature prior with a six-row published ledger (E11: six, not seven), completed against its own contract for matched comparison. Doctrine: no branch is charged on another's convention; every comparative screen must satisfy the §8.4 matched-construction conditions.

3. Branch-A device physics and erasure mechanism (E14, E16 applied)

Nominal spectrum $f01 = 0.175710013$, $f02 = 3.581724312$, $f03 = 5.207305$, $f04 = 7.982112$ GHz (3201-point grid on $[-8\pi, 8\pi]$, second-difference kinetic operator). Curvature: Richardson limit $5064.55 \text{ GHz}/\Phi_0^2$ from the four-step sequence $4426.524849 / 4873.297283 / 5014.089593 / 5051.757635$ — steps $0.020/0.010/0.005/0.0025$ are RADIANS of ϕ_{ext} , converted by $(2\pi)^2$ to GHz/Φ_0^2 (E16); full h^2 -elimination reproduces 5064.55 exactly. Desired erasure: $|2\rangle \rightarrow |1\rangle$ to the monitored sink. Unlocated bypass: the quasiparticle sine channel with $\delta \equiv \phi - \phi_{\text{ext}}$ (E14; at half flux $\sin(\delta/2) = -\cos(\phi/2)$, parity-even, coupling the same-parity pair — the loop-phase reading $\sin(\phi/2)$ is parity-odd and vanishes): $|\langle 0|\sin(\delta/2)|2\rangle| = 0.295705232$ gives $\Gamma_{\text{qp}}/x_{\text{qp}} = 4.106e10 \text{ s}^{-1}$ (180 – $210 \mu\text{eV}$ band 3.945 – $4.261e10$), making the P1 bound demanding and central. All values independently replicated to stated

precision, three builds.

4-5. Control, reset and coherence contracts (by reference)

Raman single-qubit control (P2: cycle error $\leq 1e-4$, leakage $\leq 2e-4$, N1Q ≤ 5 provisional zero-margin); FTF/tunable-coupler two-qubit gate at the p2q $\leq 8.0e-4$ ceiling (P4; record-edge comparators are context, not transferred); reset/thermal-sink map (P0d: $t_{\text{reset}} \leq 8 \mu\text{s}$, residual $|1\rangle \leq 1\%$, back-action $\leq 2.5e-4$, $\tau_{\text{th}} \geq 1.8 \text{ ms}$ — a $\geq 225\times$ switchable damping ratio on the 176 MHz sink transition); decomposed coherence (P3: $[1-W_\phi]/2 \leq 1.8e-3$; $T_\phi \geq 5.55 \text{ ms}$ at the 20 μs special case). Base §§4-5 and Appendices K/L by reference; unchanged.

6. Readout architecture: P6-S and P6-E

P6-S is syndrome/EOL readout; P6-E is the mid-circuit data-qubit erasure check, disabled in the baseline pending its sub-gates. Eq.(7) at 6.5 GHz: $(\chi_0, \chi_1, \chi_2) = (-5.366823, -0.305447, +7.214942) \text{ MHz}$ ($g/2\pi = 150 \text{ MHz}$, ten-level convention); physical pull is its negative. The operating point is the non-perturbative dressed null $\delta_0 = \delta_2$ of $H = \text{diag}(\omega_i) + \omega_r a^\dagger a + g n(a + a^\dagger)$ with adiabatic labelling; Eq.(7) supplies the seed only. At the null: pulls $-9.507/+8.721 \text{ MHz}$, contrast 18.228 MHz, sink line 4.310696 GHz, dressed $f_{12} = 3.395056 \text{ GHz}$. Fabrication tolerance: every declared-ensemble sample retains a null; the collision gate rejects $|\omega_{24} - f_r| < 13 \text{ MHz}$.

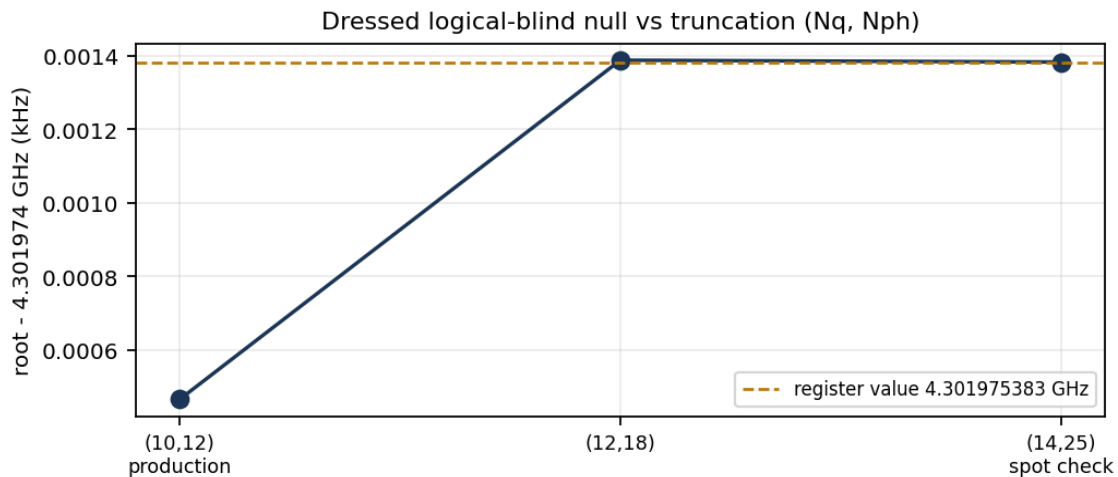


Figure 1. Dressed null vs truncation. Production (10,12) = 4.301974466 GHz; (14,25) spot check = 4.301975383 GHz = register value (E13). Data: qmhp_v158f_readout_replication.py (shipped); two independent implementations agree at all shared points.

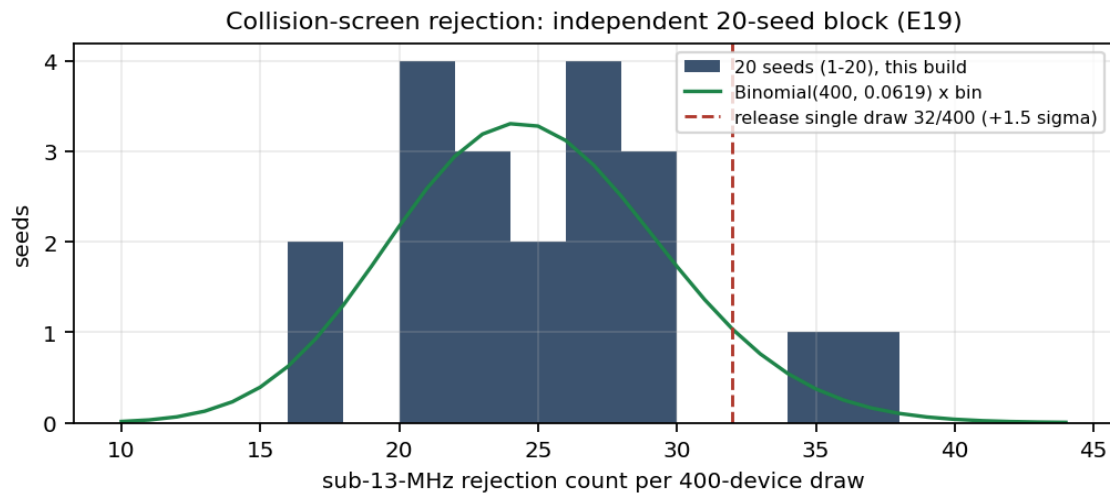


Figure 2. Collision-screen rejection, independent 20-seed block (seeds 1-20; ensemble_20seed.json, shipped): pooled 6.19% [5.66, 6.72], dispersion 1.13 vs Binomial(400, p); the release's 32/400 sits at $+1.5\sigma$. Together with the rc1 block (6.66% [6.12, 7.21]) the normative rate is $\approx 6.2\text{--}6.7\%$ (E19, ER-8).

7-8. Campaign registers and branch controls

The P6E-DRIVEN stages (dark loading; pointer dephasing; Floquet census; ideal IQ discrimination; exact Purcell channel per E18; sink-branch composition) stand as frozen. Branch-C contract: the six-row published ledger totals $1.7053\text{e-}2$; contract-complete $C_{\text{m_full}}$ totals $2.0603\text{e-}2$; time-scaled $C_{\text{m_full_ts}}$ $2.0197\text{e-}2$ (all independently reproduced exactly; the prose $\approx 1.95\text{e-}3$ of omitted classes = $1.15\text{e-}3$ additions + $0.8\text{e-}3$ correlated uplift). Contract-completion hazard cost vs C-as-published: $1.63\times / 2.37\times / 2.68\times$ at $d = 3/5/7$ (printed counts). §8.4 matched-construction conditions — same anchor kind, same surrogate, attested ledger completeness, one declared decoder policy — are pre-registration objects; any screen failing them is CONSTRUCTION-SENSITIVE and not decision-bearing.

9. Coupled error ledger and QEC-MAP results

Branch-A unlocated allocation: fourteen rows totalling $8.500\text{e-}3$ /round (re-aggregation exact); located load $P_e = 1.543\text{e-}2$ /round as twirlAD with efficiency f . Rounds: $t_A = 20\text{ }\mu\text{s}$; $t_C = 13.5\text{ }\mu\text{s}$ prior (12–15 sensitivity). Metric: $\lambda_L = -\ln(1-p_{L,\text{round}})/t_{\text{round}}$ with propagated CIs.

Screen (10 rounds, MWPM)	d=3	d=5	d=7	Status
Matched $A_{\text{m_twirlAD}} / C_{\text{m_full_ts}}$	0.94 [0.88, 1.01]	0.92 [0.79, 1.08]	1.15 [0.97, 1.36]	NO SEPARATION; different-seed Tier-C MC inside all printed CIs (published range 0.91–1.10; ER-6)
Ideal free-location bracket	0.43	0.20	0.15	A lower at $\geq 95\%$ everywhere; OPTIMISTIC upper bound on location value
Persistent-leakage LRU1 (resident sink)	$2.840\text{e-}2$	$1.835\text{e-}2$	$1.280\text{e-}2$	Sharply adverse; disputed $3.3\times$ vs v1.5.8d — P6-E7 BLOCKING
Break-even location efficiency f^*	0.524	0.598	—	Benefit-side gate of P6-E6'

Pauli-equivalence contract, schedule-sensitivity bands and the divisor-100 retirement stand as in the base (§§9.4–9.7 by reference).

10. Validation ladder and stop rules

Gate	Question	Pass evidence / threshold	Stop rule
P0d	Sink reset without damaging logical states?	$t_{\text{reset}} \leq 8\text{ }\mu\text{s}$; residual [1] $\leq 1\%$; back-action $\leq 2.5\text{e-}4$; $\tau_{\text{th}} \geq 1.8\text{ ms}$; measured rates replace anchors	No half-flux scale-up if reset/process fails
P1	Direct unlocated bypass compatible with ledger?	$\Gamma_{\text{unlocated},20} \leq 5\text{ s}^{-1}$; mechanisms project $P_u \leq 8.50\text{e-}3$ /round	Branch ends if projected $P_u > 1\text{e-}2$ /round or $\Gamma > 25\text{ s}^{-1}$
P2	Symmetry-compatible logical control?	Raman cycle error $\leq 1\text{e-}4$; leakage $\leq 2\text{e-}4$; N1Q ≤ 5 ; no control-induced QP rise	No 2Q/QEC claim without it
P3	Coherence survives the full trajectory?	$[1-W_{\phi}]/2 \leq 1.8\text{e-}3$; $T_{\phi} \geq 5.55\text{ ms}$ at $20\text{ }\mu\text{s}$; branch-resolved	Fail if averages hide below-threshold windows
P4-pre/P4	Coupler model converged; gate meets allocation?	Converged static/driven/open model; $p_{2q,\text{twirled}} \leq 8.0\text{e-}4$ with leakage/ZZ/spectators separate	No coupler fabrication / no code-tile claim otherwise
P5	Errors reliably converted to erasures?	Destination-resolved leakage/flags; monitor efficiency + recovery map	No erasure-bias claim from leakage alone
P6-S	Syndrome/EOL readout non-demolishing?	Three-state confusion matrix; dynamic MIST/Purcell/higher-mode sweep; latency closure	Redesign if back-action dominates
P6-E	Data-qubit check locates sink without damage?	P6-E1–E5 plus the P6-E6' joint contract (§10.1)	No mid-circuit reinstatement unless P6-E6' passes in full
P7-pre / QEC-MAP	Ledger maps without hidden normalization; matched?	Row-wise map re-sums; decoder frozen before holdout; equal wall clock; §8.4 on every arm	BLOCKING: no absolute p_L or selection claim otherwise
P7	Unconditional logical advantage?	$p_L < 1\%$ and $< p_{\text{ref}}$ at $\geq 95\%$ under matched construction, equal wall clock	No selection from CONSTRUCTION-SENSITIVE screens; stop/redirect A if matched measured QEC-MAP fails vs C

10.1 P6-E6' joint contract (E1-E3) and P6-E7 (E10)

P6-E6' — JOINT ACCEPTANCE OBJECT (supersedes single-object $f > f^*$; ER-5 note: all bars per data qubit per round-equivalent)

Mid-circuit checks may be reinstated only if ALL of: (1) measured effective $f \geq f^*$ ($0.524 / 0.598$ at $d=3/5$, vs C as published); (2) measured $p_{\text{induced}} \leq 1.0\text{e-}3$ — set by ledger headroom ($1.50\text{e-}3$ above the $8.50\text{e-}3$ baseline; $\approx 3.3\text{e-}4$ after the An-like $P_e \rightarrow P_u \approx 1.17\text{e-}3$ /round sink conversion), NOT by the decoder crossing ($\approx 1.1\text{--}1.3\text{e-}2$, $\approx 7.5\times$ above headroom); (3) the complete P_u ledger re-closes $\leq 1.0\text{e-}2$ /round with every enabled-check row inserted — necessary, not sufficient; the P1 stop binds independently (E2); (4) realistic-check p_L beats the frozen no-check baseline at $\geq 95\%$ at equal wall clock. A registered $5.0\text{e-}3$ bar is REJECTED: it exceeds the ceiling headroom $3.3\times$, so a pass at that bar trips K.2 and P1 simultaneously. Hardware measurement of every input remains decisive; T.12' is MODEL-DEPENDENT.

P6-E1 dressed trim (per-device root; Eq.(7) seed only); P6-E2 filter transfer ($\kappa(\omega)$ with suppression at 3.395056 GHz); P6-E3 sink survival (T_{sink} 3–6 ms class; $q_{\text{miss}} \leq 5\text{e-}4$ re-optimised jointly per E3); P6-E4 classifier; P6-E5 MIST/leakage. P6-E7 (leakage-to-syndrome coupling) is a tracked release blocker (E10): owner B. Duncan; protocol and the adjudication rule between the two LRU1 implementations frozen before comparative data exist.

11-12. Hardware build specification and change control

Per-device dressed trim in the ≈ 4.30 GHz band; mandatory 34–36 dB Purcell stopband at 3.395056 GHz (synthetic band 3.2056–3.6727 GHz, max 37.65 dB; ≥ 81.1 dB/GHz recovery; paired separation ≥ 761 MHz, median 905 MHz); coupler-band and EOL replan before tile-scale fabrication; fabrication acceptance via the ω_{24} gate at the E19 standard (≈ 5 –8% per-cooldown reject/reallocate scale); QP/IR/radiation engineering per base §11.4–11.5. Change control: frozen computation vs measured anchor; withdrawals carried forward; Appendix-S legacy tables carry the E8 watermark — PROVENANCE ONLY, superseded by Appendix T. Independent clean-room reproduction is a standing release gate (E15).

13. Risk register and decision logic

Risk	Current evidence	Decision / next gate
Direct unlocated bypass	Half-flux QP sine channel large ($\Gamma_{qp}/x_{qp} = 4.106e10 \text{ s}^{-1}$); x_{qp} diagnostic $\approx 4.6\times$ below best published	Measure schedule-level bypass per branch; stop if ledger cannot close
Reset generates QPs / damages logical states	Computationally unresolved by construction	Same-device P0d/P1, reset ON/OFF
Readout destroys sink	$q_{\text{miss}} \leq 5e-4 \Rightarrow T_{\text{sink}} 3\text{--}6 \text{ ms vs } \approx 20 \mu\text{s}$ An-like (150–300 $\times$); $P_e \rightarrow P_u \approx 1.17e-3/\text{round}$ eats $\approx 78\%$ of headroom; T.12': decoder break-even 1.1–1.3e-2, operative bound = headroom 1.5e-3 ($\approx 3.3e-4$ after conversion)	BEYOND-DEMONSTRATED. Measure $T_{\text{sink}}(n)$, destination-resolved survival; re-optimize q_{miss} jointly (E3)
Unfiltered Purcell conversion	F8 weight 0.01182198 ((12,18)/(14,25); (10,12): 0.01182201) $\Rightarrow$ microsecond [2] $\rightarrow$ [1] decay for a flat 2–3 MHz bath	Mandatory 3.3951 GHz stopband; measured $\kappa(\omega)$
Hidden Floquet resonance	[14] line bounded inert in $n = 5\text{--}8$; array/package modes not in model	EM/mode census + driven spectroscopy
Fabrication collision / trim	400/400 nulls at declared seed; rate $\approx 6.2\text{--}6.7\%$ pooled over two 20-seed blocks (E19)	Per-device calibration; reject/reallocate
Check cost destroys d=3 benefit	Ideal bracket + f*; T.12' charges model-level costs; hardware costs unmeasured	No-check baseline stands; P6-E6' governs reinstatement
Coupler/readout spectral conflict	Readout at ≈ 4.30 GHz; legacy coupler plan invalid	Replan before 2Q fabrication
HIGH: leakage-to-syndrome coupling	Two LRU1 implementations differ $3.3\times$ at d=3 and reverse the distance trend	BLOCKING; P6-E7 tracked with pre-registered adjudication (E10)
HIGH: screens are construction-sensitive	v1.5.8d ordering reverses under matched anchors/surrogates/ledgers/decoder policy; per-unit-time loads differ $\sim 6\%$	§8.4 conditions on every future screen
MODERATE: prior-edition reproducibility	S.3/S.6 do not reproduce Table S.4 ($-28\%/ -13\%$)	Remediated by the E6 channel-convention table; S.4 is provenance

Decision logic: no branch wins by literature preference and none loses by construction artifact. Two qualifications travel with that disposition. First, the distance trend: $R7/R3 = 1.221\text{--}1.223 [\approx 1.02, \approx 1.47]$, $P(R7/R3 > 1) \approx 0.986\text{--}0.987$ across independent implementations (spread ≤ 0.002 ; hazard-convention and stream-order labelled per ER-5) — EXPLORATORY, but adverse to a scaling-premised architecture. Second, T.12': the verdict turns on check-induced logical error, not location efficiency — and the ledger ceiling binds $\approx 7.5\times$ before the decoder crossing.

PREMISE-KILL WATCH (§13.2, criterion 4)

Under matched construction Branch A beats contract-complete C at zero location efficiency — a PASS on the letter, but the mechanism is A's lower charged load per unit time, not erasure conversion. If measured ledgers preserve this pattern, the erasure-specific hardware will have been carried without causing the advantage. P5 and P6-E must therefore demonstrate that location is the operative mechanism, not merely that Branch A wins. Kill criteria (1)–(4) and the pre-registration rule stand unchanged and are NOT currently triggered.

14. Appendix T' — realistic-location cost study and distance trend

14.1 T.12' model and results

Branch A carries its full $8.50\text{e-}3$ ledger plus $P_e = 1.543\text{e-}2$ as twirlAD with location efficiency f (removal convention, OPTIMISTIC). Costs per the release's own contracts: a check_induced row (p_{induced} , data, depol1) per K.1; round time $20 + t_{\text{chk}}$ μs in the hazard normalisation; the §9 SCALING box applied literally (six time/dephasing rows and P_e rescaled by $(20 + t_{\text{chk}})/20$; fixed rows at parity; fixed-round bracket alongside). Comparator: $C_{\text{m_full_ts}}$ at printed values, re-executed bit-exactly. 86 configurations, adaptive stopping to 250 failures, seeds declared (qmhp_v158f_t12_t13.py, shipped). The printed 117-build T.12 (dynamic-miss q model, 3 μs check, re-run C arm) is an independent construction of the same study; its break-even $1.22\text{e-}2$ lies inside the CIs below.

d	f	t_chk	Convention	p*_induced	95% CI
3	0.90	2 μs	scaled	$1.125\text{e-}02$	[$1.034\text{e-}02$, $1.228\text{e-}02$]
3	1.00	2 μs	scaled	$1.258\text{e-}02$	[$1.154\text{e-}02$, $1.380\text{e-}02$]
5	0.90	2 μs	scaled	$1.121\text{e-}02$	[$9.806\text{e-}03$, $1.265\text{e-}02$]
5	1.00	2 μs	scaled	$1.179\text{e-}02$	[$1.034\text{e-}02$, $1.328\text{e-}02$]
3	0.90	2 μs	fixed	$1.276\text{e-}02$	[$1.179\text{e-}02$, $1.389\text{e-}02$]
5	0.90	2 μs	fixed	$1.186\text{e-}02$	[$1.042\text{e-}02$, $1.338\text{e-}02$]
3	0.60	2 μs	scaled	$8.536\text{e-}03$	[$7.558\text{e-}03$, $9.441\text{e-}03$]

At $d = 7$ the tested points ($f = 1.00$, $p_{\text{induced}} \leq 4\text{e-}3$) remain $\geq 2.8\times$ below the matched control. F-1 (MODEL-DEPENDENT): break-even $1.1\text{--}1.3\text{e-}2$, weakly f -dependent near the ideal point, 24% tighter by $f = 0.60$ — a joint frontier; f^* retained. F-2 (NORMATIVE): the ceiling binds first — P6-E6'. F-3: Liu-like cost ($7.2\text{e-}5/\text{check}$) passes with $d=5$ HR 0.26 at $f = 0.90$; An-like ($2.49\text{e-}2/\text{check}$) exceeds the grid top (HR 1.48 at $1.6\text{e-}2$) and breaches the ledger ceiling $\approx 17\times$. The admissible band lies between two published devices, not beyond both.

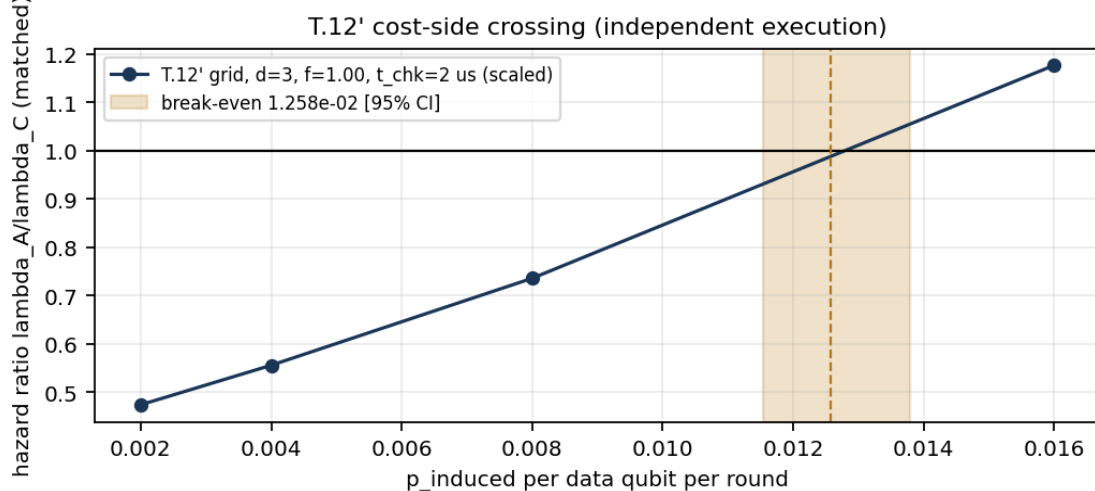


Figure 3. T.12' cost-side crossing at $d=3$, $f=1.00$ (t12_realistic_location_runs.json + t12_breakevens.json, shipped). The printed T.12 break-even $1.22\text{e-}2$ sits inside the band.

14.2 T.13' distance-trend statistic

$R7/R3 = \text{HR}(d=7)/\text{HR}(d=3)$ on the released S1 counts. Release convention ($-\ln(1-p)/t$ hazard), seed 13, 20,000 resamples: 1.2217 [1.0206 , 1.4632], $P(R7/R3 > 1) = 0.986$. Small- p linear-hazard convention, same seed: 1.223 [1.025 , 1.465], $P = 0.987$. Printed 117-build value (seed 11, 200,000): 1.221 [1.022 , 1.461], $P = 0.986$. Spread across conventions and streams ≤ 0.003 (ER-5 labels applied). Legs: $R5/R3 = 0.981$ [0.826 , 1.166] (flat); $R7/R5 = 1.247$ [0.993 , 1.568], $P = 0.972$ (not separated): the adverse move is concentrated at the $d = 7$ endpoint. Shipped artifacts: release-hazard primary and linear sensitivity JSONs, convention-named (RB-3). Status: RECOMPUTATION-FROM-RUN-RECORD / EXPLORATORY; frozen-procedure pre-registration recorded for the next campaign (E7).

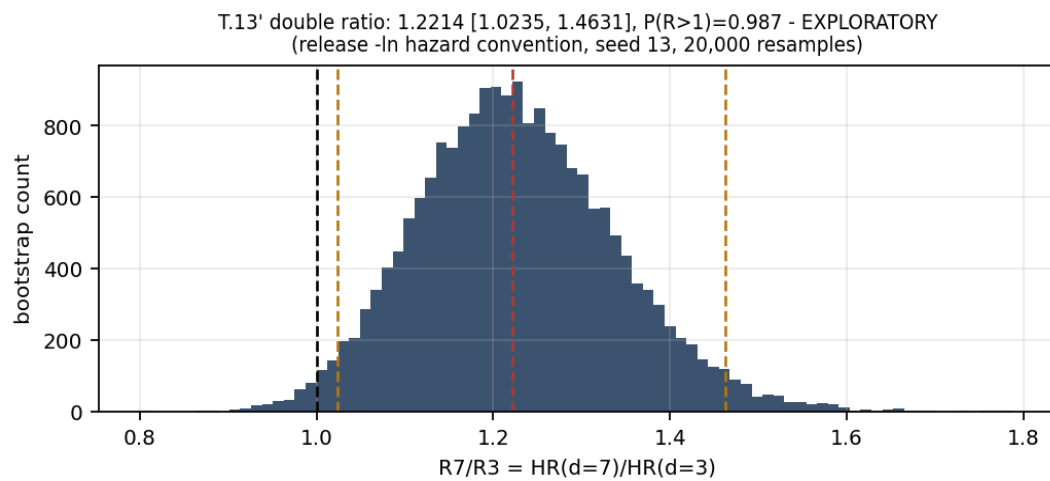


Figure 4. T.13' bootstrap (t13_distance_trend.json, shipped; release hazard convention). Carries no confirmatory weight.

15. Consolidated verification register

Layer	Status	Evidence
Static Hamiltonian / spectrum / matrix elements	INDEPENDENTLY REPLICATED	Nine figures; two Tier-C builds; n01/n03/n12/n23/n25 to six figures
Curvature + Richardson; QP bypass (H.4/H.5)	INDEPENDENTLY REPLICATED	Four radian-step rows 8-9 s.f.; limit 5064.55 exact under h^2 -elimination; $4.106e10 \text{ s}^{-1}$
Eq.(7) diagnostic; perturbative root	INDEPENDENTLY REPLICATED	χ triple exact; 4.314628047 GHz nine figures
Exact dressed null + truncation provenance	STRONGLY VERIFIED	(10,12)/(12,18)/(14,25) ladder; E13
Pulls / sink line / dressed f12	INDEPENDENTLY REPLICATED	-9.507/+8.721; 4.310696; 3.395056 GHz (truncation-stable)
Purcell weight (F8)	RESOLVED — release exact	0.01182198 at (12,18)/(14,25); (10,12) 0.01182201; bare-admixture 0.011815 identified (E18)
Ensemble rejection rate	STATISTICALLY SUPPORTED	≈ 6.2 - 6.7% across two 20-seed blocks; dispersion binomial (E19)
Exact ensemble counts / percentile digits	UNDER-DETERMINED BY PROSE SPEC	Exact once the stream→(EJ,EC,EL) order is declared — demonstrated by the Tier-B 32/400 + all-percentile reproduction (ER-7)
Ledger / row-wise map / marginals / C totals	INDEPENDENTLY REPLICATED	14 rows <1e-9; 8.500e-3; 1.7053/2.0603/2.0197e-2 exact
Statistics layer (39 rows)	INDEPENDENTLY REPLICATED	pL/λ/Wilson reproduce from raw counts
Stochastic QEC (matched/free/LRU1)	INDEPENDENTLY SUPPORTED	Tier-B bit-exact (29/39) + different-seed Tier-C inside printed CIs
T.12' break-even	REPLICATED WITHIN CI	Printed 1.22e-2 inside blind-execution CIs; conventions differ and are declared
T.13' distance trend	INDEPENDENTLY SUPPORTED (EXPLORATORY)	≤ 0.003 spread across three implementations
Hardware validity	OPEN — decision-bearing	P0d-P7 gates; §10

16. Edit register E1-E20 (APPLIED)

E1 P6-E6' joint contract replaces single-object $f > f^*$. E2 sub-gate pass necessary-not-sufficient; Pu re-closure and P1 bind independently. E3 T.12' numbers into the risk register; joint q_{miss} re-optimisation. E4 realistic-location variant executed at model level; hardware decisive. E5 S6 corrected (model-level check costs exist in S2.2/T.12). E6 channel-convention table travels with every screen. E7 T.13' EXPLORATORY + pre-registration. E8 legacy Appendix-S watermark. E9 decision brief (front matter). E10 P6-E7 tracked blocker with owner and ordering gate. E11 six-row comment corrected. E12 $1.8\times$ withdrawn; executed per-axis ratio $4/3$ (empirical 1.231 at $d=3$). E13 dressed-null truncation label. E14 $\delta \equiv \phi - \phi_{\text{ext}}$ convention printed (§3). E15 standing independent-reproduction release gate. E16 curvature step-units caption (§3). E17 two-build designation: 117-page operative, 109-page provenance, both hashes in the manifest. E18 Purcell F8 adjudication with truncation label (ER-4). E19 ensemble reporting standard: multi-seed rate normative; exact counts adjudicable only with a declared draw order (ER-7). E20 cross-implementation agreement wording with hazard-convention labels (ER-5). All twenty applied inline; AMD-C r6 + ERRATA-1 remain the change-record artifacts. All defects found in the August 2026 verification cycle are registered and closed; none is decision-bearing.

17. Conclusion and final program position

The row-wise QEC-MAP method survives and is strengthened; the branch conclusion built on the v1.5.8d construction does not. Charging both branches at the same anchor kind, with the same twirled-amplitude-damping surrogate, on contract-complete ledgers, under one declared decoder policy, with the location bracket present, leaves the branches statistically indistinguishable at every tested distance. Against a properly-charged C, Branch A wins at zero location efficiency — the margin comes from lower charged load per unit time, not erasure conversion, and the premise-kill watch applies. The persistent-leakage variant remains sharply adverse and its coupling strength is the most model-sensitive number in the program. The burden has shifted to construction discipline and then to hardware.

FINAL PROGRAM POSITION

Proceed to fabrication/measurement only as a gated experiment. This edition supports building a device to test the architecture. It does not support a processor-performance or fault-tolerance claim, and it does not support the withdrawn v1.5.8d conclusion that the erasure branch has been shown not to earn its hardware. The next evidence increment is hardware — P6-E7 and the P0d/P1 vehicle first — and the next methodological requirement is matched construction.

18. Artifact manifest

Artifact	SHA-256
v1.5.8e 117-page build — OPERATIVE base (E17)	d334cda822c07c212be1fac850a34debef5ea66ee45899e56b44e19154f86c6c
v1.5.8e 109-page render — provenance	83aeb3a2d8ebcb8f627989557350d9c172ad5bc20c3edc561b8a5e091cc4b658
qmhp_v158f_t12_t13.py	1474e3f34c11e1e875392dd19e7b10ed2a0f42e1577e75ee2418397ebe8e9ed3
qmhp_v158f_readout_replication.py	ba8486b19d34b61cd2ccff9e0bd76b53c348d1834f40f8c5ffcc21ad8880d490
build_master_figures.py	888ae2a39315706cf7193334da16df9ef1181235e0cc8384fee28ecfa506f9c3
t12_realistic_location_runs.json	abab6054faae342d6ddeed9702866dc868c354eebb3f4b0b6e8ee02428c10e86

Artifact	SHA-256
t12_breakevens.json	282a52afbcbb98202bd7a9719652538dd3615a6deca8ab596803da39ec54f410
t13_distance_trend_release_hazard.json	d55c0db17c5d63e3a6337f61787decfa57415fce71f9e0e3affb8ccff13f5130
t13_distance_trend_linear_sensitivity.json	4c207a8243b81d82927d2aa338928cf98f726a56e2cd854d51d96f1baaf6ff27
ensemble_20seed.json	3f755149d175eed1686d6f2a2ad5186ad7587b3064de96743ba6d15241828965
qmhp_v158e_runs.json	174977acbf09e02064715dba970644e1d7e8d3340068d21f3a5a287f2ce1c9c0
figM1_truncation.png	c8adaaf10e10e22b4724cb695879c0750d8ec5d9771503cb1ee785ee3ca5b540
figM2_ensemble.png	b22e1340e4eff29985c8ef3436c90efc256055cf29c2bd6f781feaaab6553654
figM3_t12_crossing.png	d1fc2a4ff1a27b5a3cd81c777f2deec16a066d690b1c4da392889bd2e120bc39
figM4_t13.png	bc031885975fff440384068b9dbaf02a78a0492f108aa814aad08f1919db1630

AMD-C r6 (change record), ERRATA-1 and the full reproduction bundle (RUN.md, requirements.lock, expected_outputs.json, reconstructed release modules) are carried under their published digests. Fixed-point rule unchanged: this document cannot contain its own digest; it is written to the external release manifest after rendering.